

Ex 2

(1) SELECT DISTINCT departement
FROM Student;

Departement
IT
HR
Finance

(2) SELECT departement, AVG(age) AS avg-age
FROM Student;

Departement	AVG-age
IT	20.5
HR	22.0
Finance	23.0

3 SELECT departement, COUNT(*) AS Student count
FROM Students
Group by departement
Having COUNT(*) > 1;

Departement	Student count
IT	2
HR	2

(4) SELECT Student id, name, age, departement
FROM Students
WHERE age BETWEEN 21 AND 28;

Student Id	Name	Age	departement
3	Charlie	21	IT
2	Bob	22	IT
4	Diana	23	Finance
5	Eve	22	HR

5 SELECT Student Id, name, age, departement
 FROM employees
 WHERE departement 'IT' OR 'HR'
 AND Age > 21

Student Id	name	age	departement
2	Bob	22	HR
5	Eve	22	HR

6 ~~SELECT departement, Sum(Credits) AS total Credit
 FROM Courses
 GROUP BY departement
 Having (Credits) > 5;~~

$$3 + 4 = 7 + 4$$

$$11$$

SELECT departement, Sum(Credits) AS total Credits
 FROM Courses
 GROUP BY departement
 Having Sum(total Credits) > 5;

departement	Total Credits
IT	11

7 SELECT course_id, course name, departement, credits
 FROM COURSES
 WHERE Credits < > 4

Course Id	Course name	departement	Credits
101	SQL Basic	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8 SELECT course id, Course name, credits
 FROM enrollments
~~GROUP ORDER~~

3 SELECT course id, course name, credits
 FROM enrollments course
 ORDER By credits DESC
 Limit 3 ;

Course Id	Course name	Credits
102	Python	4
103	Data Science	4
104	SQL Basic	3

~~SELECT Min(*) AS min-grade, Max(*) AS max grade,
 AVG(*) AS Avg-grade
 FROM enrollments~~

min-grade	Max grade	avg grade

19 SELECT Max(grade) AS Max-grade,
Min(grade) AS Min-grade
Avg(grade) AS Avg-grade
FROM enrollments;

Max grade	90
Min-grade	78
Avg grade	84.6

10 SELECT course_id,

COUNT(*) AS enrollment counts
FROM enrollment;
Group By course_id;

course_id	enrollment count
101	1
102	1
103	1
104	1
105	1

11 SELECT department,

SUM(salary) AS total salary,
SUM(bonus)
FROM salaries
Group by department;

Department	total salary	total bonus
IT	129,000	10500
HR	109,000	7560
Finance	70,000	6600

12 SELECT departement,
 AVG(Salary) AS avg-salary
 FROM salaries
 Group By departement
 Having Avg(salaries) > 55.000;

departement	avg salary
IT	61000
Finance	70000

13 ~~SELECT~~ employee id, name, salary + bonus
 FROM salaries
 WHERE salary + bonus > 600

SELECT employee id, name, salary, bonus,
 bonus + salary AS total-compensation
 From salaries
 WHERE salary + bonus > 60000

Employee id	name	salary	bonus	total compensation
1	Tom	60000	5000	65000
4	Tyke	62000	5500	67500
3	Spike	70000	6000	76000

14 SELECT department
 SUM(budget) AS total-budget, ~~Exercise~~
 AVG(budget) AS avg-budget
 FROM Project
 Having Avg(budget) > 70000;

Department	total budget	avg budget
IT	270000	185000
Finance	80000	80000

15 ~~SELECT~~ ~~Project_id~~ ~~Project_name~~ ~~Department~~ ~~budget~~

15 SELECT Project_id, Project_name, Department
 , budget
 FROM Projects
 WHERE budget BETWEEN 50000 AND 120000
 AND Department <> 'Marketing';

Project id	Project_name	Department	budget
1	AI APP	IT	120000
2	Payroll System	Finance	80000
5	HR Portal	HR	50000